## Jishnu Nambissan Thekkeppattu (he/him)


**Contact Information**

CSIRO Space and Astronomy,
26 Pembroke Road,
Marsfield, NSW 2122, Australia.
jishnunambissan.thekkeppattu@csiro.au
https://jnthek.github.io/


**Research Interests**

Radio astronomy, radio interferometry, instrumentation and techniques. Phased arrays, calibration and array signal processing, wide-field polarimetry. High speed DSP, reconfigurable computing, FPGAs. Experimental and observational 21-cm cosmology, precision radiometry, data analysis for 21-cm experiments, radio-sky modelling at low-frequencies. All-sky spherical harmonic transit interferometry ($m$-mode imaging). Numerical linear algebra, ill-posed inverse problems, iterative methods. Ionospheric scintillation.

**Professional experience**

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| Aug 2025 - Present | CERC Fellow, CSIRO Space and Astronomy, Australia |
| Aug 2022 - Jul 2025 | Research Associate, Curtin University, Australia |
| Nov 2016 - Jul 2022 | Research Scientist/Research Associate, Raman Research Institute, India |
| Jan 2016 - Nov 2016 | Assistant System Engineer, Tata Consultancy Services, India |

**Education**

**Curtin University, Australia**

Doctor of Philosophy – Physics (2019–2022)

- Thesis title : Towards detection of redshifted 21-cm signal from Cosmic Dawn and Epoch of Reionization.
- Thesis supervisors : Dr. Cathryn M. Trott, Dr. Benjamin McKinley (Curtin Institute of Radio Astronomy, Bentley, WA)

**Kannur University, India**

Bachelor of Technology in Electronics and Communication Engineering (2011–2015)

- Aggregate percentage : 81.2%
- Undergraduate project : A VHF board for Software Defined Radio (SDR) applications.
- Project advisor: Dr. V. Vinodkumar (Govt. College of Engineering, Kannur)

**Publications**


ORCID : 0000-0002-3079-4046


**First authored, peer reviewed, journal papers**

**Jishnu N. Thekkeppattu**, Randall Wayth, and Marcin Sokolowski, *Calibration of an SKA-Low prototype station using holographic techniques*, 2024, Radio Science https://doi.org/10.1029/2023RS007847.

**Jishnu N. Thekkeppattu**, Cathryn M. Trott, and Benjamin McKinley, *Singular spectrum analysis of time series data from low frequency radiometers, with an application to SITARA data*, 2023, Monthly Notices of the Royal Astronomical Society, https://doi.org/10.1093/mnras/stad522.

**Jishnu N. Thekkeppattu**, Benjamin McKinley, Cathryn M. Trott, Jake Jones, Daniel C. X. Ung, *System design and calibration of SITARA - a global 21-cm short spacing interferometer prototype*, 2022, Publications of the Astronomical Society of Australia, https://doi.org/10.1017/pasa.2022.13

**Jishnu Nambissan, T.**; Ravi Subrahmanyan; R. Somashekar; N. Udaya Shankar; Saurabh Singh; A. Raghunathan; B. S. Girish; K. S. Srivani; Mayuri Sathyanarayana Rao *SARAS 3 CD/EoR radiometer: design and performance of the receiver*, 2021, Experimental Astronomy, https://doi.org/10.1007/s10686-020-09697-2

**Co-authored, peer reviewed, journal papers**

Macario, Giulia; Pupillo, Giuseppe; Bernardi, Gianni; Di Ninni, Paola; Comoretto, Giovanni; Bolli, Pietro; Mattana, Andrea; Monari, Jader; Perini, Federico; Schiaffino, Marco; Sokolowski, Marcin; Wayth, Randall; **Thekkeppattu, Jishnu N.**, *Intrinsic cross-polarization ratio maps from all-sky observations with the SKA-Low prototype station Aperture Array Verification System 2*, Journal of Astronomical Telescopes, Instruments, and Systems, Volume 11, id. 028008 (2025)
https://doi.org/10.1117/1.JATIS.11.2.028008

Saurabh Singh, **Jishnu Nambissan T.**, Ravi Subrahmanyan, N. Udaya Shankar, B. S. Girish, A. Raghunathan, R. Somashekar, K. S. Srivani, Mayuri Sathyanarayana Rao, *On the detection of a cosmic dawn signal in the radio background*, Nature Astronomy, 6, 607-617 (2022) https://doi.org/10.1038/s41550-022-01610-5

Girish, B. S.; Srivani, K. S.; Subrahmanyan, Ravi; Udaya Shankar, N.; Singh, Saurabh; **Jishnu Nambissan, T.**; Sathyanarayana Rao, Mayuri; Somashekar, R.; Raghunathan, A. *SARAS CD/EoR Radiometer: Design and Performance of the Digital Correlation Spectrometer*, 2020, Journal of Astronomical Instrumentation, Volume 9, Issue 2. https://doi.org/10.1142/S2251171720500063

B McKinley, C M Trott, M Sokolowski, R B Wayth, A Sutinjo, N Patra, **J. Nambissan T.**, D C X Ung, The All-Sky SignAl Short-Spacing INterferometer (ASSASSIN) - I. Global-sky measurements with the Engineering Development Array-2, Monthly Notices of the Royal Astronomical Society, Volume 499, Issue 1, November 2020, Pages 52-67, https://doi.org/10.1093/mnras/staa2804

Raghunathan, A.; Subrahmanyan, R; Shankar, N. U.; Singh, S; **Nambissan, J.**; Kavitha, K.; Mahesh, N.; Somashekar, R.; Sindhu, G; Girish, B. S.; Srivani, K. S.; Rao, M. S., *A Floating Octave Bandwidth Cone-Disk Antenna for Detection of Cosmic Dawn*, in IEEE Transactions on Antennas and Propagation, vol. 69, no. 10, pp. 6209-6217, Oct. 2021, https://doi.org/10.1109/TAP.2021.3069563

Adrian T. Sutinjo, Benjamin McKinley, Leonid Belostotski, Daniel C. X. Ung, and **Jishnu Nambissan T.**, *Design Equations for Closely Spaced Two-Element Interferometer for Radio Cosmology*, URSI Radio Science Letters, Vol. 2, 2020,
https://doi.org/10.46620/20-0017

**Conference Proceedings**

**Thekkeppattu, J. N.** (2026). *Monitoring equatorial ionospheric scintillation using low-cost software defined radios - some results from the Indian sector.* In Proceedings of the XXXVIth URSI General Assembly and Scientific Symposium - URSI GASS 2026. XXXVIth URSI General Assembly and Scientific Symposium. URSI - International Union of Radio Science.
https://doi.org/10.46620/URSIGASS26/IVTK6571

**Thekkeppattu, J. N.**, Humphrey, D., Thyagarajan, N. (2026). *Ultrafast real-time imaging with large-N aperture arrays using FPGA accelerators.* In Proceedings of

the XXXVIth URSI General Assembly and Scientific Symposium - URSI GASS 2026. XXXVIth URSI General Assembly and Scientific Symposium. URSI - International Union of Radio Science.
https://doi.org/10.46620/URSIGASS26/CUUS2167

Thyagarajan, N., **Thekkeppattu, J. N.**, Humphrey, D.(2026). *A fast, wide-field, and real-time imaging prototype for large-aperture arrays.* In Proc. SPIE 14153, Radio Telescopes, Technologies, and Methods, 141530G.
https://doi.org/10.1117/12.3105286

**Thekkeppattu, J. N.**, Cook, J. H., Wayth, R. B., Carozzi, T. D., Sokolowski, M., Kriele, M. A., and Vijayaraghavan, M. (2025). *All-sky m-mode map making at 159 MHz with EDA2 and SE607HBA LOFAR station.* In Proceedings of the 7th URSI Asia-Pacific Radio Science Conference - AP-RASC 2025. 7th URSI Asia-Pacific RadioScience Conference. URSI - International Union of Radio Science.
https://doi.org/10.46620/ursiaprasc25/wqun6574

Cook, J., **Thekkeppattu, J.**, Wayth, R., and Kriele, M. (2025). *All Sky maps from m-mode Spherical Harmonic Transit Interferometry with the MWA and the EDA2.* In Proceedings of the 7th URSI Asia-Pacific Radio Science Conference - AP-RASC 2025. 7th URSI Asia-Pacific Radio Science Conference. URSI - International Union of Radio Science.
https://doi.org/10.46620/ursiaprasc25/pomb8661

**Thekkeppattu, J. N.**, Sokolowski, M., and Wayth, R. B. (2025). *Broadband self-holography of EDA2 - an SKA-Low prototype station.* In Proceedings of the 7th URSI Asia-Pacific RadioScience Conference - AP-RASC 2025. 7th URSI Asia-Pacific Radio Science Conference. URSI - International Union of Radio Science.
https://doi.org/10.46620/ursiaprasc25/hech4944

Raghunathan, Agaram; Girish, B.S.; Somashekar, R.; Srivani, K.S.; Singh, Saurabh; Subrahmanyan, Ravi; Shankar, N. Udaya; Rao, Mayuri Sathyanarayana; **Nambissan T., Jishnu**, *Wide band antenna with Ultra-smooth spectral characteristics*, 2020 International Applied Computational Electromagnetics Society Symposium (ACES), 2020, pp. 1-2, https://doi.org/10.23919/ACES49320.2020.9196075

**Nambissan, T. J.**, Nikhil, T., Vinodkumar, V., *A VHF Radio for Software Defined Radio Applications*, 2016, Procedia Technology, 24, 820, International Conference on Emerging Trends in Engineering, Science and Technology (ICETEST - 2015)
https://doi.org/10.1016/j.protcy.2016.05.109

**Non peer reviewed articles / pre-prints**

**Jishnu N. Thekkeppattu**, *First results from a low cost software defined radio system monitoring VHF equatorial ionospheric scintillations in the Indian sector*, 2024, ESS open archive.

## Research Experience

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| 2025-Present | Development of an ultrafast radio camera using FPGAs<br>Supervisor(s): Dr. Nithyanandan Thyagarajan, Dr. David Humphrey<br>CSIRO Space and Astronomy, Marsfield, NSW. |
| 2022-2025 | SKA-Low bridging and Early Production Array activities &<br>All-sky mapping with spherical harmonic transit interferometry<br>Supervisor(s): Dr. Randall Wayth, Dr. David Davidson<br>Curtin Institute of Radio Astronomy, Bentley, WA. |
| 2019–2022 | Towards detection of redshifted 21-cm signal from Cosmic Dawn and Epoch of Reionization.<br>Supervisor(s): Dr. Cathryn Trott, Dr. Benjamin McKinley,<br>Curtin Institute of Radio Astronomy, Bentley, WA. |
| 2019–2021 | Principal investigator of PRATUSH: Probing ReionizATion of the Universe using Signal from Hydrogen.<br>(Funded for pre-project studies by the ISRO)<br>Co-PIs : Dr. Saurabh Singh, Dr. Mayuri S. Rao<br>Raman Research Institute, Bangalore, India. |
| 2016–2021 | On the detection of global 21-cm signal from Cosmic Dawn and Epoch of Reionization.<br>Advisors: Dr. Ravi Subrahmanyan, Dr. N. Udaya Shankar,<br>Raman Research Institute, Bangalore, India. |
| 2019 June–August | Comparing LOFAR Epoch of Reionization target fields using 21-cm power spectra.<br>Advisor : Dr. André Offringa,<br>ASTRON, Dwingeloo, The Netherlands. |
| 2014–2015 | Reconfigurable radios using Field Programmable Gate Arrays.<br>Advisor: Dr. V. Vinodkumar,<br>Govt. College of Engineering, Kannur, India |
| 2014–2015 | A VHF board for Software Defined Radio (SDR) applications.<br>Advisor: Dr. V. Vinodkumar,<br>Govt. College of Engineering, Kannur, India |
| 2013–2014 | Multiplierless filter realisation using Canonic Signed Digits.<br>Advisor: Dr. Manju Manuel,<br>Govt. College of Engineering, Kannur, India |

## Selected talks

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| 2026 Aug | *Monitoring equatorial ionospheric scintillation using low-cost software defined radios - some results from the Indian sector* - URSI-GASS-2026, held at Krakow, Poland (Young Scientist Award talk) |
| 2026 Aug | *Ultrafast real-time imaging with large-N aperture arrays using FPGA accelerators* - URSI-GASS-2026, held at Krakow, Poland |
| 2025 Dec | *Ultra-fast real-time imaging with large aperture arrays* - SALF-XI, held at Orleans, France (remote talk) |
| 2025 Aug | *All-sky m-mode map making at 159 MHz with EDA2 and SE607HBA LOFAR station* - URSI-AP-RASC 2025, held at Sydney, Australia |
| 2025 Aug | *Broadband self-holography of EDA2 - an SKA-Low prototype station* - URSI-AP-RASC 2025, held at Sydney, Australia |
| 2025 Jun | *All-sky map making with spherical harmonics in the SKA-Low era* - SKAO General Science Meeting, A new era in astrophysics: Preparing for early science with the SKAO, held at Goerlitz, Germany |
| 2020 Jan | *Updates from the SARAS experiment* - International School And Conference On Observing The First Billion Years Of The Universe Using The Next Generation Telescopes, held at IIT Indore, India (remote talk) |

## Honors and Awards

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| 2026 | Young Scientist Award, URSI-GASS-2026 |
| 2022 | Joint winner of the ICRAR Ken and Julie Michael Prize |
| 2014 | Best student paper award in embedded systems and VLSI category at National Conference on Systems Energy and Environment - 14 held at Govt College of Engineering, Kannur |
| 2010 | Indian Space Research Organisation (ISRO) sponsored student delegate, 97$^{th}$ Indian Science Congress held at Kerala University. |

## Teaching

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| 2025 | COMT4003 - Mobile Radio Communications; Unit Co-ordinator & Lecturer (Curtin University) |
| 2024 | COMT6005 - Satellite Communications; Lecturer (Curtin University) |
| 2024 | COMT4003 - Mobile Radio Communications; Lecturer (Curtin University) |
| 2023 | COMT6005 - Satellite Communications; Lecturer (Curtin University) |
| 2020 | ELEN3002/6017 - Engineering Electromagnetic Labs (Curtin University) |

## Memberships

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| 2023-Present | Junior member - International Astronomical Union (IAU) |
| 2024-2025 | Member - Institute of Electrical and Electronics Engineers (IEEE) |
| 2013-2014 | Student member - IEEE |

## Software tools and packages

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| Programming Languages: | python, C, C++, `bash` scripting |
| HDL: | VHDL |
| Software packages: | AMD Vivado, MATLAB, `miriad` |
| Operating systems: | Linux, Mac OS |
| Misc: | x86 and 8051 assembly, embedded C, arduino, Visual C# Verilog, EM simulators (NEC2, CST) and HPC systems |

## Observing and Field Experience

Deployment, operation and maintenance of a prototype two-element broadband interferometer SITARA at Inyarrimanha Ilgari Bundara - the CSIRO Murchison Radio-astronomy Observatory (2021-2022).

Measurement of global sky brightness temperature at low frequencies deploying SARAS at remote sites at Timbaktu Collective, India (2017-18), Indian Astronomical Observatory, Ladakh, India (2017-18) and Mavathur lake, India (2019).

Electromagnetic tests and measurements of performance and characteristics of the SARAS antennas and receivers at the Gauribidanur Field Station, India (2017-19) and Mavathur lake, India (2019).

## Hardware and laboratory experience

Expertise in electronics and RF/microwave systems; using vector network analysers (Keysight FieldFox, PNA-X etc.) and their calibration, spectrum analysers, system level RF design, development and testing. Hands-on experience in handling fibre-optics, optical power measurements and RFoF systems. Experienced in performing measurements in an electromagnetic compatibility (EMC) chamber. Expertise in basic laboratory/workshop tasks such as soldering and miscellaneous mechanical works. FPGA systems (Zynq, Alveo v80).

## Outreach

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| 2025–Present | CSIRO S&A Marsfield colloquia organiser. |
| 2020–2025 | DIY radio telescope programme. In collaboration with the Curtin ASTRO-3D node, built a low-cost hobby radio telescope operating at 1420 MHz for outreach purposes. |
| 2015 | Institute of Electrical and Electronics Engineers - Teachers In Service programme (IEEE-TISP) volunteer. Conducted science and engineering outreach programmes for high school students. |

## Volunteering and Hobbies

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| 2021-Present | Australian Red Cross Lifeblood blood, platelets & plasma donor |
| 2014-Present | Amateur radio operator, holding Australian callsign VK6JN and Indian callsign VU3VWB |